

Assignment 3

SIL7165: Network Systems and Security

Network Traffic Analysis

Security Threat Detection Report

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Overview

This report summarises the analysis of network traffic logs from a firewall/IDS system. The dataset consists of 6 CSV files containing 2,071,657 flows captured over 7 days (June 11–18, 2010). Each flow has 21 attributes including IPs, ports, protocol, byte/packet counts, timestamps, and a ground-truth label (Normal: 96.7%, Attack: 3.3%).

All 6 files were combined for a unified analysis since they represent the same network across consecutive days. Files were processed in chunks of 100,000 rows to handle the 1.5 GB dataset without loading everything into memory at once. The complete code and outputs are in the accompanying Jupyter notebook `assignment3.ipynb`.

Phase 1 — Basic Network Traffic Statistics

The first phase establishes a baseline picture of the network. TCP dominates at 79.4% of flows, followed by UDP at 20.2%. A small number of rare protocols (igmp: 77 flows, ip: 66 flows) were flagged as potentially suspicious.

The average packet size is 316.8 bytes but the median is only 195.8 bytes. This gap indicates a right-skewed distribution — a few very large bulk-transfer flows pull the mean upward. The median is a more reliable measure here. Variance is high ($\approx 71,150$ bytes²) due to the coexistence of tiny control packets (≈ 40 – 60 bytes) and large data transfers.

Using total bytes sent as the metric, **192.168.2.107** immediately stands out — it sent 1,766 MB, roughly 5 times more than the second-ranked host. This makes it the primary suspect from Phase 1 onward.

Hourly spike detection (threshold = mean + 3σ = 53,530 flows/hour) found two spikes: June 13 at 67,952 flows/hr ($z = 4.10$) and June 15 at 63,821 flows/hr ($z = 3.78$). Both are confirmed attack events in Phase 4. 25 internal IPs were consistently active across 99.3% of observed hours, indicating always-on servers or compromised machines.

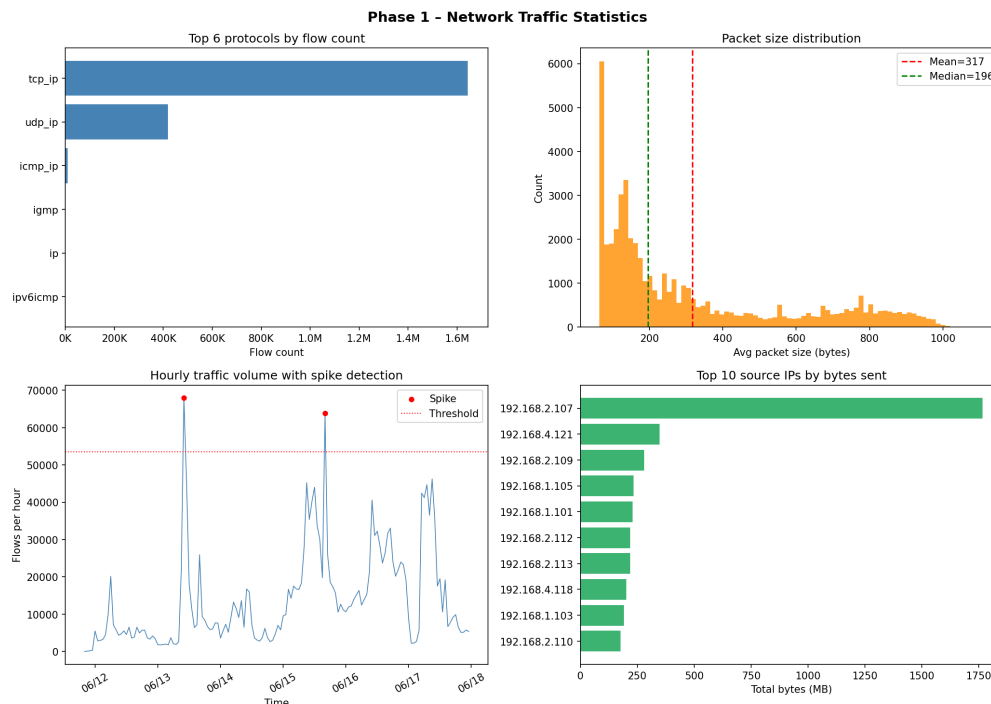


Figure 1: Phase 1: Protocol distribution, packet size histogram, hourly traffic with spike detection, top source IPs by bytes sent.

Phase 2 — Sublinear Space Algorithms

Real-world network monitors cannot store complete state for every IP. This phase implements three space-efficient algorithms that give approximate answers using a fraction of the memory required by exact methods. All three stayed within the required 10% error budget.

HyperLogLog — Counting Unique IPs

HyperLogLog uses 4,096 registers (4 KB total) and a hash-based leading-zeros trick to estimate cardinality without storing individual IPs. It estimated 35,805 unique IPs against the exact count of 34,801 — an error of 2.88% using 595 times less memory than an exact Python set.

Count-Min Sketch — Finding Heavy Hitters

A fixed 7×2000 counter array (109 KB) replaces a dictionary that would grow proportionally with input size. Each IP is hashed to 7 positions and queries return the minimum to suppress collisions. The maximum error across the top 20 destination IPs was 0.64%, using 24 times less memory than an exact dictionary.

Bloom Filter — Membership Testing

A 2.9 KB bit array with 6 hash functions answers “has this IP been seen?” with zero false negatives (a mathematical guarantee) and 1.12% false positives. This uses 58 times less memory than an exact set. A pre-pass counted the 2,478 unique source IPs so the filter could be sized correctly before insertion.

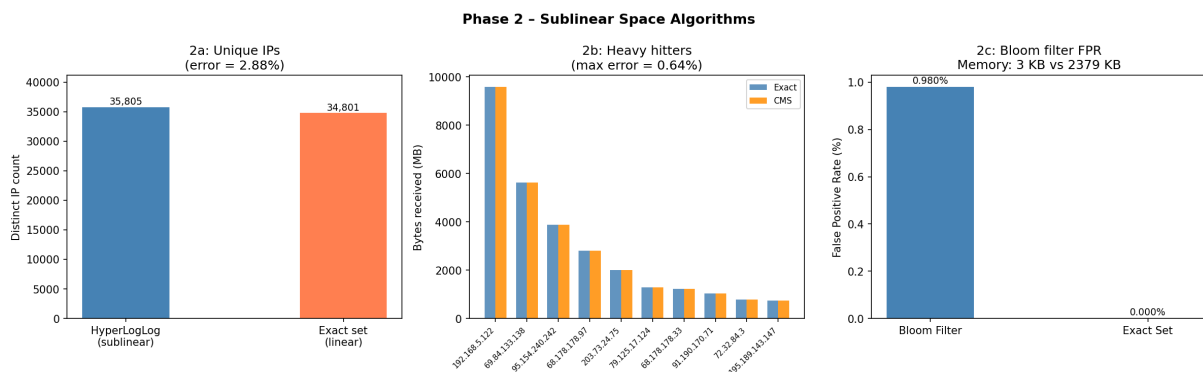


Figure 2: Phase 2: HLL vs exact cardinality, CMS vs exact heavy hitters, Bloom filter memory and error comparison.

Phase 3 — Anomaly Detection

Statistical Anomalies

Using mean + 3σ as the anomaly threshold, 21 source IPs were flagged for unusually high flow counts. The most extreme were 192.168.5.122 at 268,267 flows ($z = 26.6$) and 192.168.2.107 at 208,379 flows ($z = 20.7$). A z -score above 10 is essentially impossible under normal conditions. Ten IPs also showed anomalous byte volumes, led by 192.168.2.109 at 7,369 MB ($z = 20.1$).

Hourly analysis has a higher coefficient of variation ($CV = 0.92$) than daily analysis ($CV = 0.71$), confirming that attacks appear as short bursts that daily aggregation would smooth over and miss.

Behavioural Anomalies

192.168.2.107 received traffic from 468 distinct source IPs in a single hour on June 15, against a normal average of 1.36 sources per destination per hour. Across the full dataset, 2,345 unique sources contacted this IP ($z = 181.7$). This is either a sustained DDoS attack against this host, or this host is a C2 server being contacted by its botnet.

One IP (131.202.243.90) was silent for 46 hours and then suddenly sent 5,116 flows in one hour — consistent with a dormant bot activating.

Suspicious Patterns

7,628 flows exceeded the long-duration threshold of 1,908 seconds (31.8 min). The longest ran for 49,140 seconds (13.6 hours) from 192.168.1.101 to the broadcast address 255.255.255.255. Several other internal hosts joined this same broadcast pattern within minutes on June 17 — a strong indicator of worm propagation.

Six internal IPs used three or more protocols within a 10-minute window (TCP + UDP + ICMP or IGMP), consistent with network reconnaissance scanning.

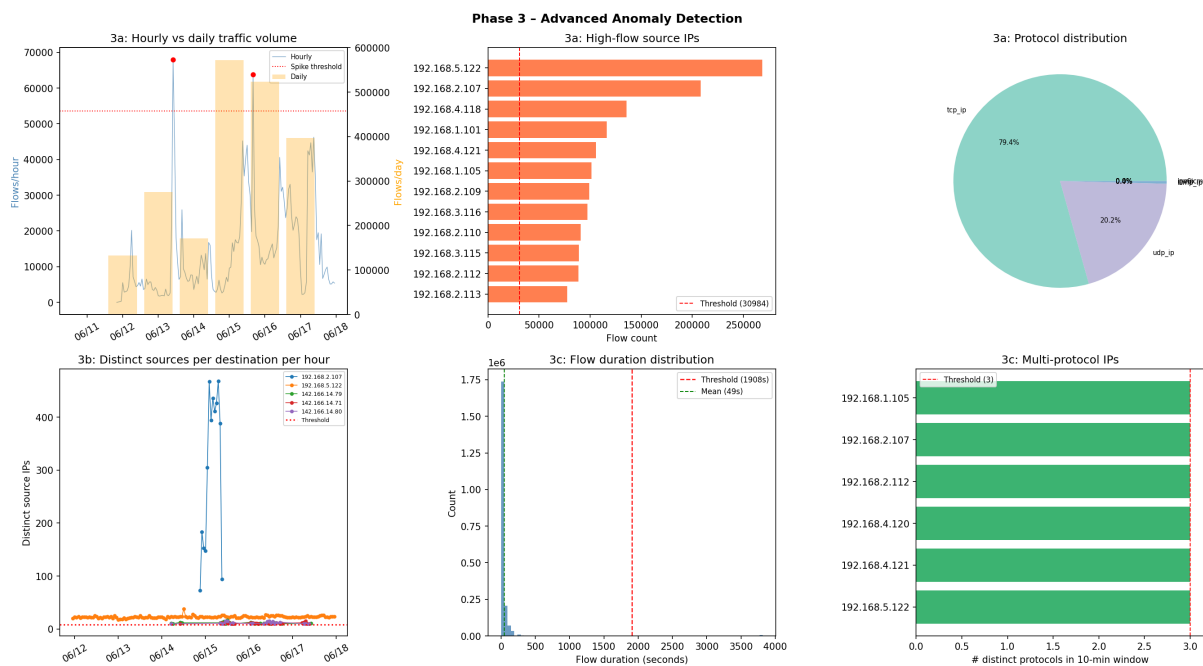


Figure 3: Phase 3: High-flow IPs, byte anomalies, hourly/daily comparison, DDoS source counts per hour, flow duration distribution, multi-protocol detection.

Phase 4 — Deep Threat Analysis

Attack Patterns

26 hosts were identified as stealthy port scanners (more than 15 distinct ports probed, spread over more than 30 minutes). The slow spread is key — it distinguishes deliberate evasion from fast obvious scans that IDS rules already catch. 192.168.2.107 probed 24,894 distinct ports across 23,744 destinations over the entire 144-hour observation window.

778 destinations showed a slow DDoS pattern — sustained hourly flow growth of at least 10% for 4 or more consecutive hours with no sharp spike. 198.164.30.2 grew 3,791% over 4 hours.

82.98.86.183 grew 52,520% over 9 hours. The gradual ramp-up is precisely what standard threshold-based IDS systems miss.

1,159 IP hopping patterns were detected where five or more IPs from the same /24 subnet all contacted the same destination. The entire 192.168.2.0/24 subnet (hosts .106 to .113) contacted hundreds of external IPs in a coordinated pattern — a compromised subnet acting as a botnet.

Payload and C2 Analysis

494,419 flows exceeded 10,000 bytes (23.9% of all flows), of which 7.6% are labelled as attacks. The largest single flow was 1,281.86 MB. 192.168.2.107 transferred approximately 798 MB to the internal host 192.168.5.122 in two separate flows, suggesting internal data exfiltration.

192.168.2.112 made repeated 4–8 MB transfers on port 443. Normal HTTPS browsing stays well below 100 KB per flow, so these are consistent with data exfiltration tunnelled through encrypted channels.

The most definitive finding is the C2 beaconing from 192.168.2.107 to 239.192.152.143: 136 flows with a mean inter-arrival time of exactly 300 seconds and a coefficient of variation of 0.000. Human browsing is irregular with CV well above 1.0. A CV of zero means every beacon arrived at exactly the 5-minute mark — this is automated, confirmed malware.

Risk Classification

839 threats were classified: **28 HIGH**, **20 MEDIUM**, and **791 LOW**. HIGH includes port scanners probing over 1,000 ports, single flows over 100 MB, and C2 beaconing with CV below 0.05. MEDIUM covers slow DDoS peaks above 1,000 flows/hr and mildly periodic communication. LOW covers minor probing and early-stage traffic growth.

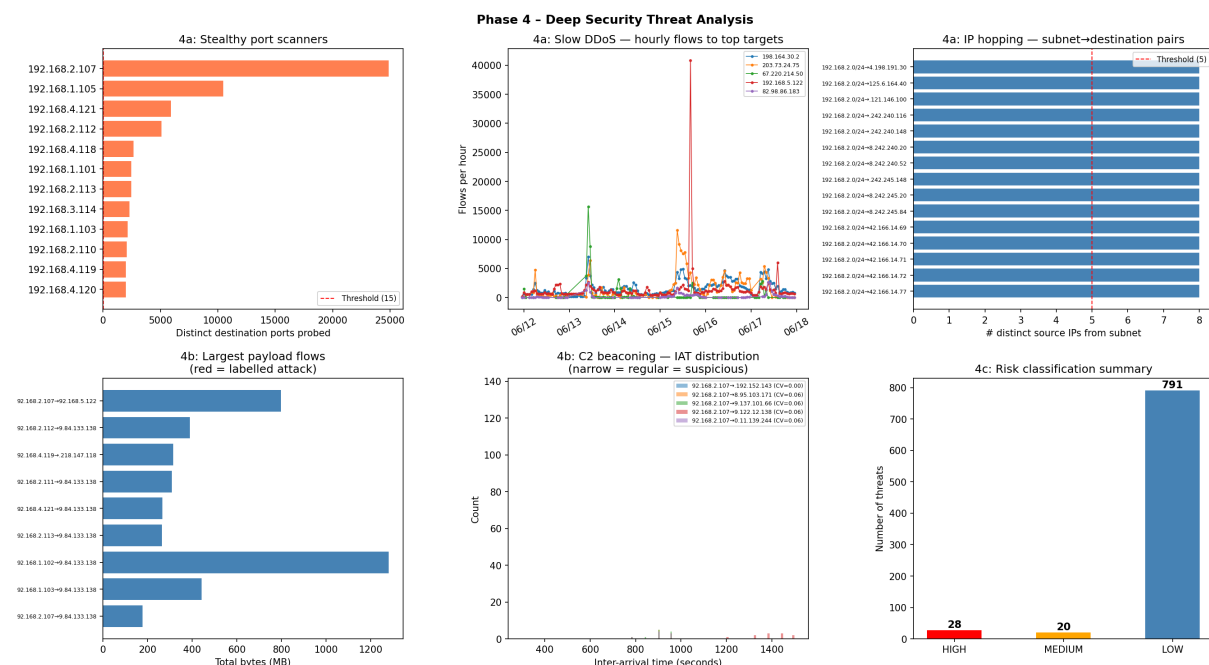


Figure 4: Phase 4: Port scanners, slow DDoS timelines, IP hopping patterns, largest flows, C2 beaconing IAT distributions, risk summary.

Key Findings and Recommendations

192.168.2.107 is the central finding of this analysis. It was simultaneously a port scanner, a C2 server, a DDoS node, and a data exfiltration source — all within the same 7-day window. The machine-precise beaconing at exactly 300-second intervals ($CV = 0.000$) confirms active malware.

1. **Isolate 192.168.2.107 immediately.** Confirmed C2 node with active malware infection. Highest-priority incident response target.
2. **Investigate the June 17 broadcast event.** Multiple hosts opened 13-hour connections to 255.255.255.255 simultaneously — signature of a self-propagating worm.
3. **Block all 65 beaconing destination IPs** at the perimeter firewall, especially the multicast C2 address 239.192.152.143.
4. **Audit the 192.168.2.0/24 subnet.** All 8 hosts contacted 1,159 external destinations in a coordinated pattern — the subnet is likely acting as an internal botnet.
5. **Inspect HTTPS traffic from 192.168.2.112.** Repeated 4–8 MB flows on port 443 are consistent with exfiltration over encrypted channels.
6. **Deploy rate-limiting** on any internal host exceeding 30,000 flows per hour.

Despite attacks comprising only 3.3% of flows, they produced 839 identified threat events. Detecting them required combining statistical thresholds, sublinear space algorithms, and behavioural analysis — no single technique alone would have uncovered the full picture.